Sortly

Task 1 - Planning - Amelia Mangion IDM 5.2

1. Project Overview

Introduction

Sortly is a responsive, database-driven web application designed to simplify household waste management in Malta. The platform aims to provide residents with a centralised and accessible service where they can check waste collection schedules, search for correct waste disposal methods, locate recycling facilities and access information about waste-related public services.

The website will be developed using HTML, CSS, JavaScript, PHP and MySQL. PHP will handle server-side functionality, including user authentication, database queries and content management, while MySQL will store structured information such as user accounts, waste items, disposal instructions, localities and collection schedules.

Sortly will support two main types of users: general residents and administrators. Residents will be able to browse waste information, create accounts, save their locality preferences and access personalised collection information. Administrators will be responsible for managing the website's waste database, disposal instructions and other relevant content.

The platform will be accessible through desktop, tablet and mobile browsers, with an emphasis on responsive design, straightforward navigation and accessibility.

The overall objective is to develop a functional web application that demonstrates database integration, account management, CRUD operations and dynamic content generation while addressing a practical problem affecting residents in Malta.

Feature 1 – Account Management

Sortly will include a user account management system allowing residents to register, log in and manage their personal information.

Functional requirements:

- Users can register using their name, email address and password.
- Registered users can log in and log out securely.
- Users can edit their name, email address, password and preferred locality.
- The system will distinguish between general users and administrators through role-based access control.
- User information will be stored in the MySQL database.
- Passwords will be securely hashed before being stored.

Non-functional requirements:

- Account forms must be easy to understand and accessible.

- Passwords must never be stored as plain text.
- User sessions must be managed securely.
- The system must prevent unauthorised access to administrative functionality.

Feature 2 – Locality-Based Collection Calendar

The collection calendar will allow users to view waste collection schedules relevant to their selected locality.

Functional requirements:

- Users can select a Maltese locality from a dropdown menu.
- The website will retrieve collection schedules from the database.
- The calendar will display collection days for organic, recyclable, mixed and glass waste.
- Each waste category will be visually distinguished using labels and colours.
- Registered users can save their preferred locality.
- Administrators can create, update and remove collection schedule records.

Non-functional requirements:

- Calendar information must be presented clearly on mobile and desktop devices.
- Colour must not be the only method used to identify waste categories.
- The calendar should load efficiently and remain easy to navigate.
- Dates and collection information must be displayed consistently.

Feature 3 – Next Collection Display

The homepage will feature a personalised section showing the next scheduled waste collection for the user's selected locality.

Functional requirements:

- The system will determine the next collection using the current date and the relevant database records.
- The homepage will display the waste category and scheduled collection date.
- Users can change their selected locality.
- The displayed information will update when a different locality is selected.
- If no collection is scheduled, the system will display an appropriate message.

Non-functional requirements:

- The information must be clearly visible without requiring extensive scrolling.
- The system must calculate the next collection accurately.
- The display must adapt to different screen sizes.

Feature 4 – Waste Disposal Search and Item Management

Sortly will include a searchable database of household waste items, allowing users to identify the correct disposal method for specific materials.

This feature will serve as the application's primary item management system.

Functional requirements:

- Users can search for waste items using keywords.
- Users can browse a list of available waste items.
- Users can select an item to open its individual detail page.
- Each item will contain a name, description, waste category, and relevant disposal information.
- Administrators can create new waste items.
- Administrators can edit existing waste items.
- Administrators can delete waste items.
- Search results will be retrieved dynamically from MySQL.
- The system will display an appropriate message when no matching items are found.

Non-functional requirements:

- Search functionality should return relevant results efficiently.
- The search interface must be straightforward and accessible.
- User input must be validated to protect database integrity.
- Database queries must use prepared statements to reduce SQL injection risks.

Feature 5 – Disposal Instruction Management

Each waste item will have associated disposal instructions explaining how the material should be handled.

This will serve as Sortly's secondary item management system, directly linked to the primary waste items.

Functional requirements:

- Each waste item can have one or more disposal instructions.
- Instructions will include a description and an optional display order.
- Administrators can create new disposal instructions.
- Administrators can edit existing instructions.
- Administrators can delete instructions.
- Instructions will be displayed dynamically on the relevant waste item's detail page.
- Each instruction will be associated with a valid waste item through a database relationship.

Non-functional requirements:

- Instructions must be clearly organised and easy to read.
- Database relationships must prevent orphaned instruction records.
- Administrative forms must validate submitted information.
- The interface must display instructions consistently across devices.

Feature 6 – Recycling Centre Locator

Sortly will provide a recycling centre directory to help residents identify facilities where specific waste materials can be disposed of.

Functional requirements:

- Users can browse available recycling facilities.
- Each facility will have a dedicated information section displaying its name, address, locality and accepted waste types.
- Users can filter facilities by locality or accepted material.
- A map may be integrated to display facility locations.
- Administrators can manage recycling facility information through the database.

Non-functional requirements:

- Facility information must be clearly presented.
- Location details must be accurate and maintained.
- The directory must be usable on mobile and desktop devices.
- If a map is implemented, a text-based directory must remain available as an accessible alternative.

Feature 7 – Waste Management Guidance

The website will include an educational section providing guidance on waste separation and responsible disposal practices in Malta.

Functional requirements:

- Users can browse informational pages covering different waste categories.
- The website will display instructions explaining what materials belong in each category.
- Relevant waste items can link to additional guidance.
- Administrators can update informational content when waste management guidelines change.

Non-functional requirements:

- Information must be presented in clear, understandable language.
- Content must be organised using appropriate headings and sections.
- Guidance should be based on reliable waste management sources.
- The website must maintain a consistent visual layout.

Feature 8 – Waste Fines and Online Payment Information

Sortly will provide an informational section explaining waste-related fines and directing users to relevant official payment services.

Functional requirements:

- Users can browse information about waste disposal regulations and associated penalties.
- The website will display relevant fine descriptions and official reference links.
- Where an authorised online payment service exists, users can follow a link to the official platform.

- Administrators can update fine-related information and links.

Non-functional requirements:

- Legal and penalty information must be based on authoritative sources.
- The website must clearly distinguish information from official payment services.
- Sortly will not collect payment card information or process fines directly.
- Content must be reviewed when relevant regulations change.

Feature 9 – Responsive Interface and Saved Preferences

Sortly will be designed as a responsive web application, allowing users to access its features through different devices.

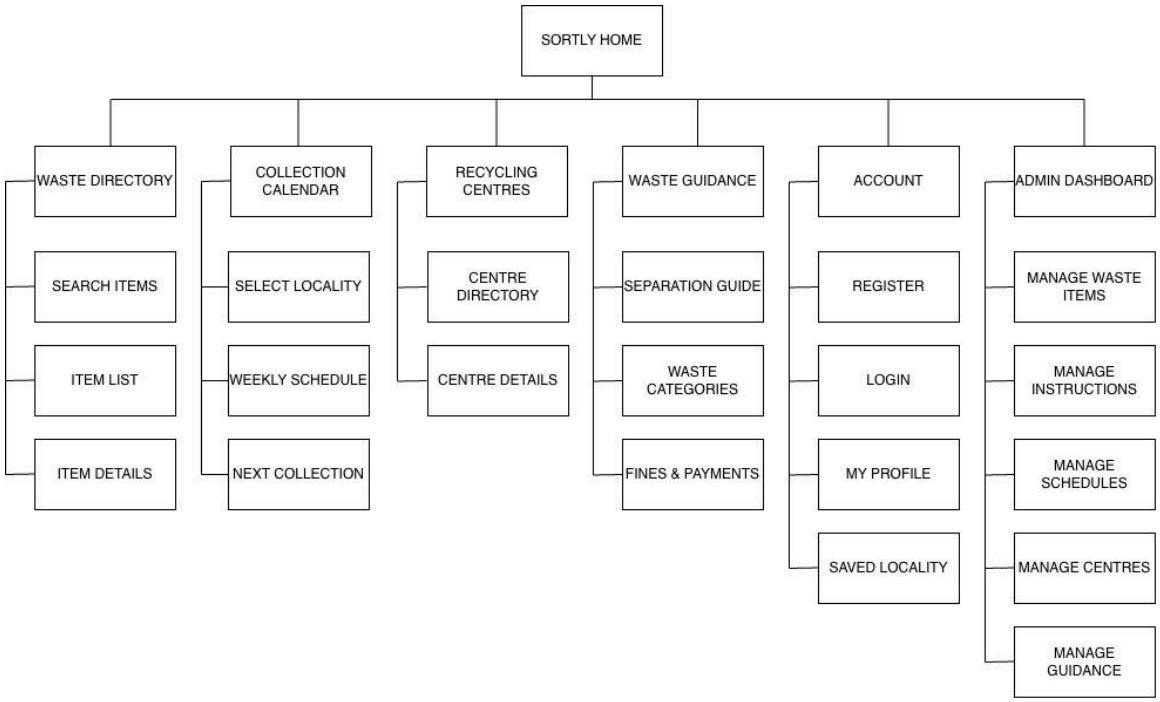
Functional requirements:

- Users can select their preferred locality.
- The selected locality can be saved using browser storage for visitors.
- Registered users can store their locality preference in the database.
- The website will retrieve saved preferences when users return.
- Users can change their preferences at any time.

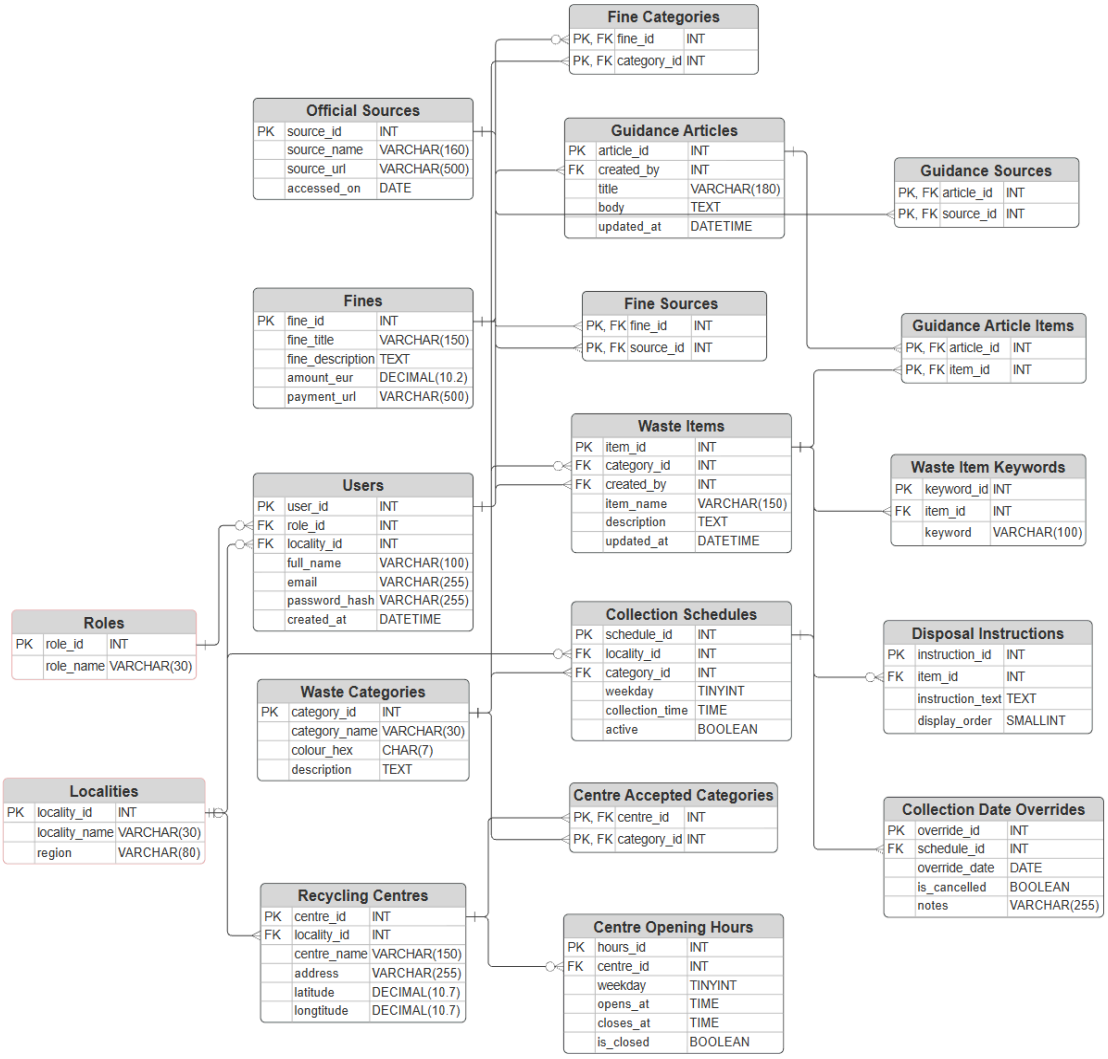
Non-functional requirements:

- The interface must adapt to mobile, tablet and desktop screen sizes.
- Navigation must remain consistent throughout the website.
- The design should follow WCAG 2.2 accessibility principles.
- Text must remain readable at different viewport sizes.
- Interactive elements must support keyboard navigation.
- The website should provide appropriate feedback when errors occur.

2. Sitemap

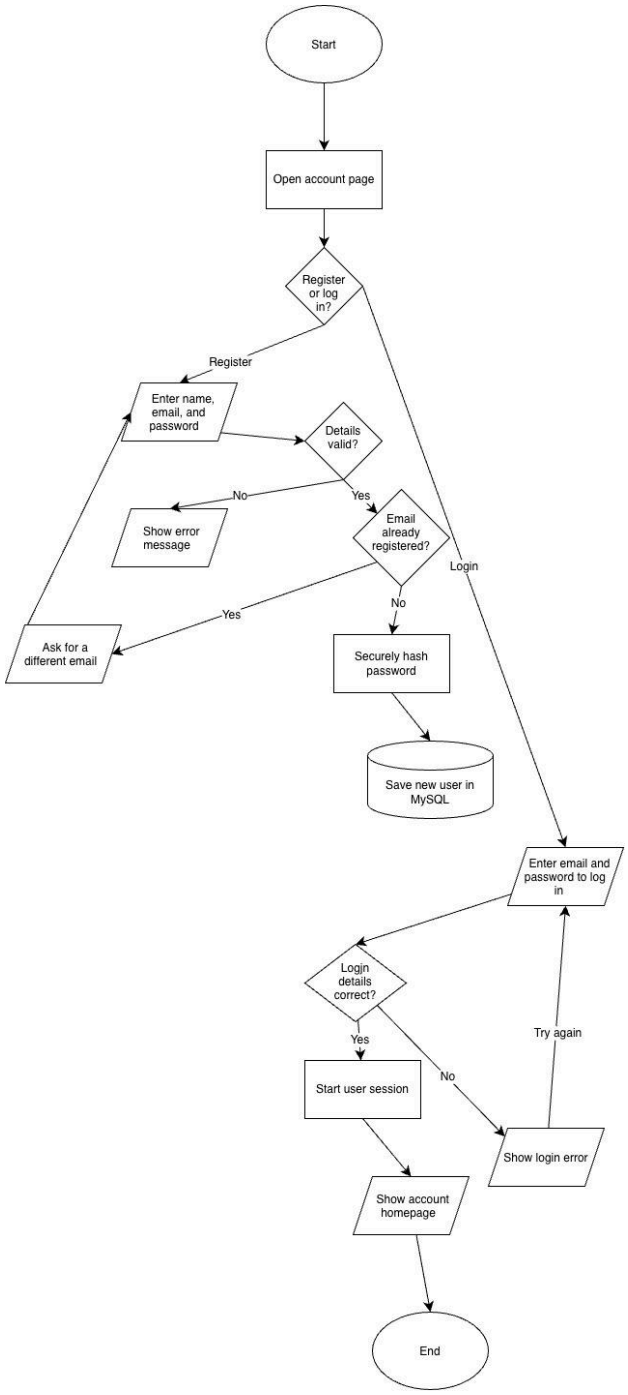

SORTLY HOME
WASTE DIRECTORY
SEARCH ITEMS
ITEM LIST
ITEM DETAILS
COLLECTION CALENDAR
SELECT LOCALITY
WEEKLY SCHEDULE
NEXT COLLECTION
RECYCLING CENTRES
CENTRE DIRECTORY
CENTRE DETAILS
WASTE GUIDANCE
SEPARATION GUIDE
WASTE CATEGORIES
FINES & PAYMENTS
ACCOUNT
REGISTER
LOGIN
MY PROFILE
SAVED LOCALITY
ADMIN DASHBOARD
MANAGE WASTE ITEMS
MANAGE INSTRUCTIONS
MANAGE SCHEDULES
MANAGE CENTRES
MANAGE GUIDANCE

3. ERD

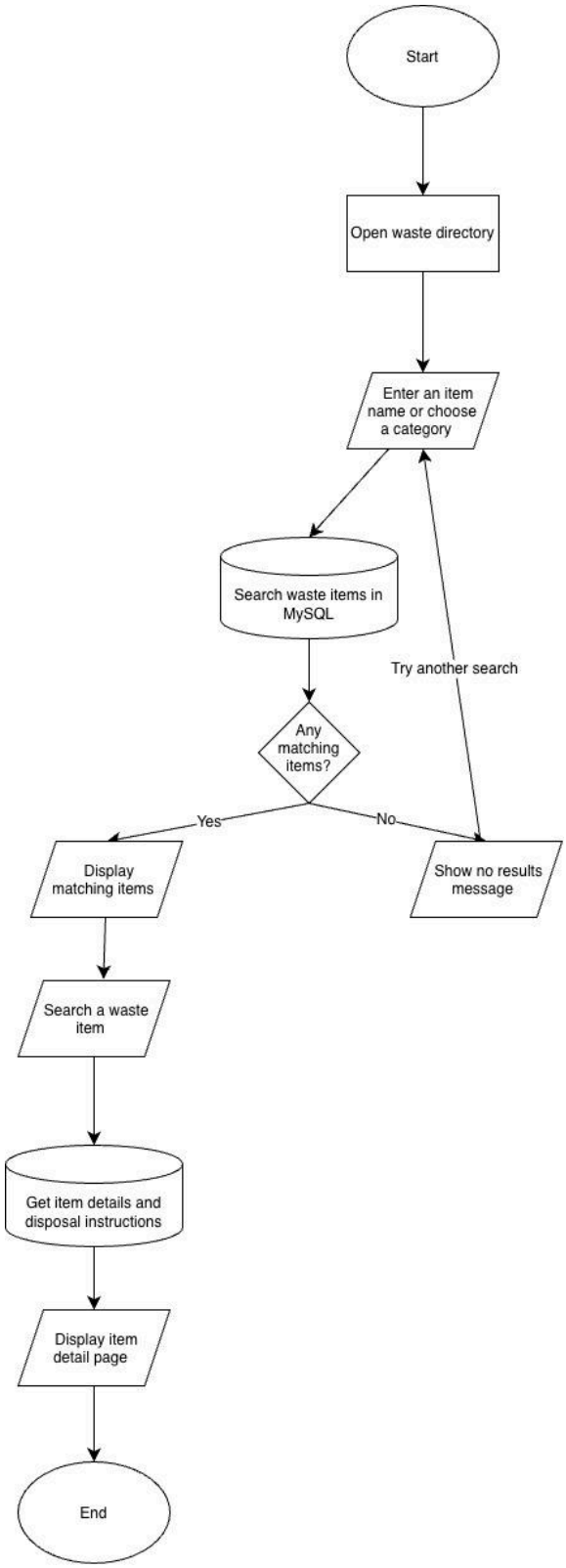


4. Flow Charts

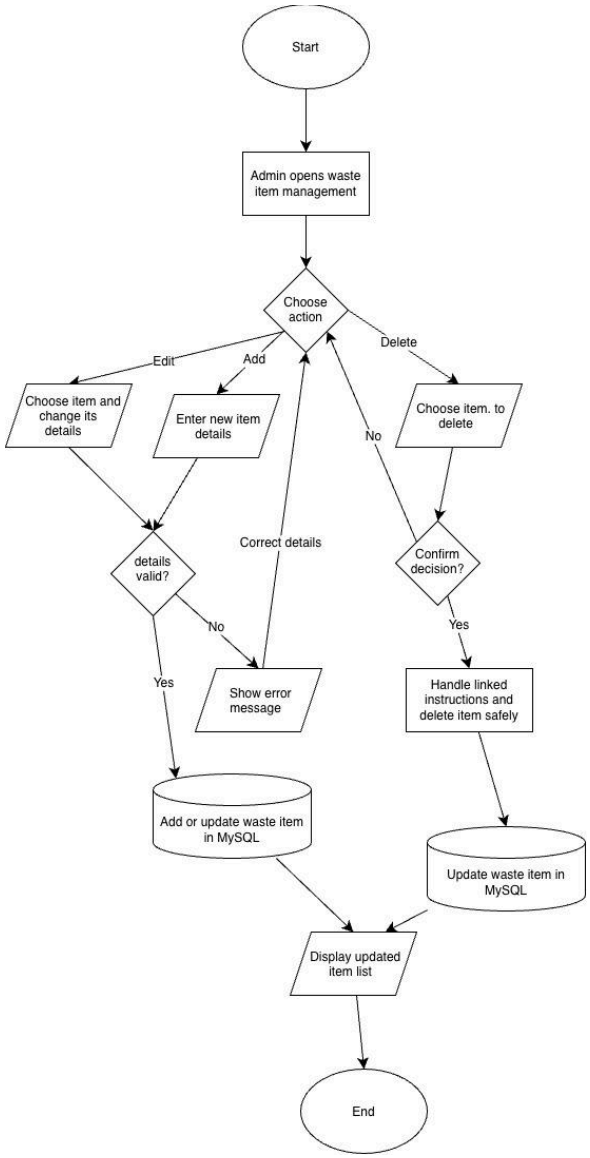
1. Account registration & login



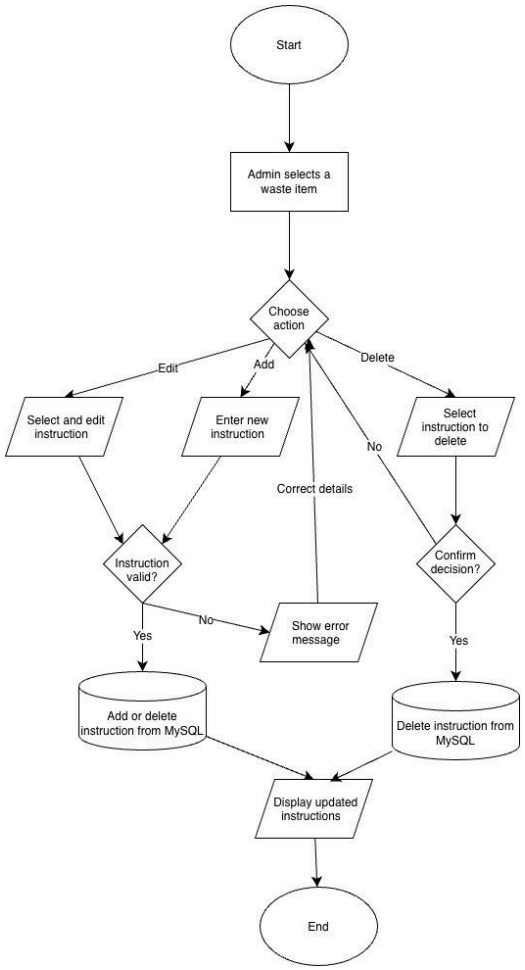
2. Waste item search



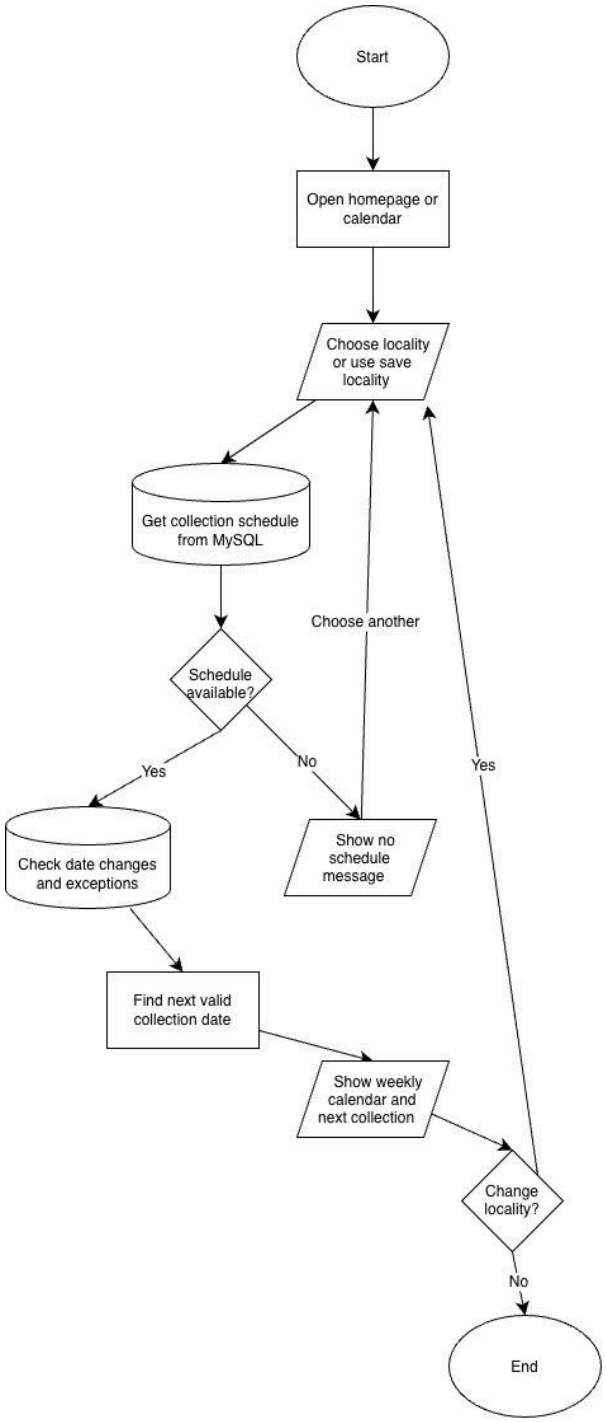
3. Waste item management



4. Disposal instructions



5. Collection calendar and next collection



5. IPO Charts

1. User Registration

Input	Process	Output
Name, email and password	Check that all details are filled in correctly	Account created or error shown
Email already in use	Check if email is already registered	Ask user to use another email

Password	Save password safely	Password stored securely

2. Login and Logout

Input	Process	Output
Email and password	Check details against saved account	User logs in or sees an error
Logout button	End the user session	User is logged out

3. Edit Account Details

Input	Process	Output
New name or email	Check details and save changes	Account details updated
New password	Check and save new password securely	Password updated
Preferred locality	Save selected locality	Locality appears in user account

4. Collection Calendar

Input	Process	Output
Selected locality	Find collection days for that locality	Weekly collection calendar shown
Selected date	Check for changes to the usual schedule	Correct collection information shown
Locality with no schedule	Check if information is available	Message shown explaining no schedule is available

5. Next Collection

Input	Process	Output
Current date and locality	Find the next scheduled collection day	Next date and waste type shown
Change of locality	Find collection details for the new locality	Next collection display updates
No upcoming collection found	Check if there is a matching collection	Helpful message shown

6. Waste Item Search

Input	Process	Output
Item name or keyword	Search saved waste items	Matching items shown
Selected waste category	Show only items in that category	Filtered list shown
Click on an item	Find full details of the item	Waste item details page opens
Search with no results	Check whether any items match	No results message shown

7. Manage Waste Items (Main Items)

Input	Process	Output
New waste item details	Check and save the new item	Item added to the list
Changes to an existing item	Find and update the item	Updated item shown
Delete item request	Check permission and remove item safely	Item removed or explanation shown
View items	Load saved waste items	Waste item list displayed

8. Manage Disposal Instructions (Related Items)

Input	Process	Output

New instruction and waste item	Check that the waste item exists and save instruction	Instruction added to item
Edited instruction	Save the changes	Updated instruction shown
Delete instruction request	Remove selected instruction	Instruction removed
Open waste item page	Find instructions for that item	Instructions displayed in order

9. Recycling Centre Directory

Input	Process	Output
Open directory	Find recycling centres	List of centres shown
Selected locality	Show centres in that locality	Filtered centres shown
Selected waste type	Find centres that accept that waste	Suitable centres shown
Click a centre	Find its address, opening times and accepted waste	Centre details shown

10. Manage Recycling Centres

Input	Process	Output
New centre details	Check and save the details	New centre added
Edited centre details	Save updated information	Centre details updated
Accepted waste types	Save which waste types the centre accepts	Accepted types shown
Opening times	Save days and times	Opening times shown
Delete centre request	Remove centre and its related information safely	Centre removed

11. Waste Guidance

Input	Process	Output
Open guidance page	Find saved guidance articles	Articles displayed
Choose an article	Find its text and related waste items	Full article shown
New or edited article	Check and save the article	Article added or updated
Delete article request	Remove the article safely	Article removed
Official source link	Find the source for the article	Source link displayed

12. Waste Fines and Payment Information

Input	Process	Output
Open fines page	Find saved fine information	List of fines shown
Choose a fine	Find its description and amount	Fine details displayed
Official payment link	Check the saved link	User can open official payment website
Admin changes fine details	Check and save changes	Fine information updated

13. Saved Locality and Responsive Layout

Input	Process	Output
Visitor selects locality	Save choice in the browser	Locality remembered on return
Logged-in user selects locality	Save choice in the account	Locality available after login
User returns to website	Load saved locality	Personalised information shown

Phone, tablet or desktop screen	Adjust page layout to screen size	Website fits the screen

14. Administrator Access

Input	Process	Output
User opens admin page	Check whether user is an administrator	Admin page opens or access denied
Resident tries to change content	Check account permission	Change blocked
Admin saves content	Check permission and details	Changes saved
User is logged out	Check login status	User asked to log in